

R&B: Radon Bridge

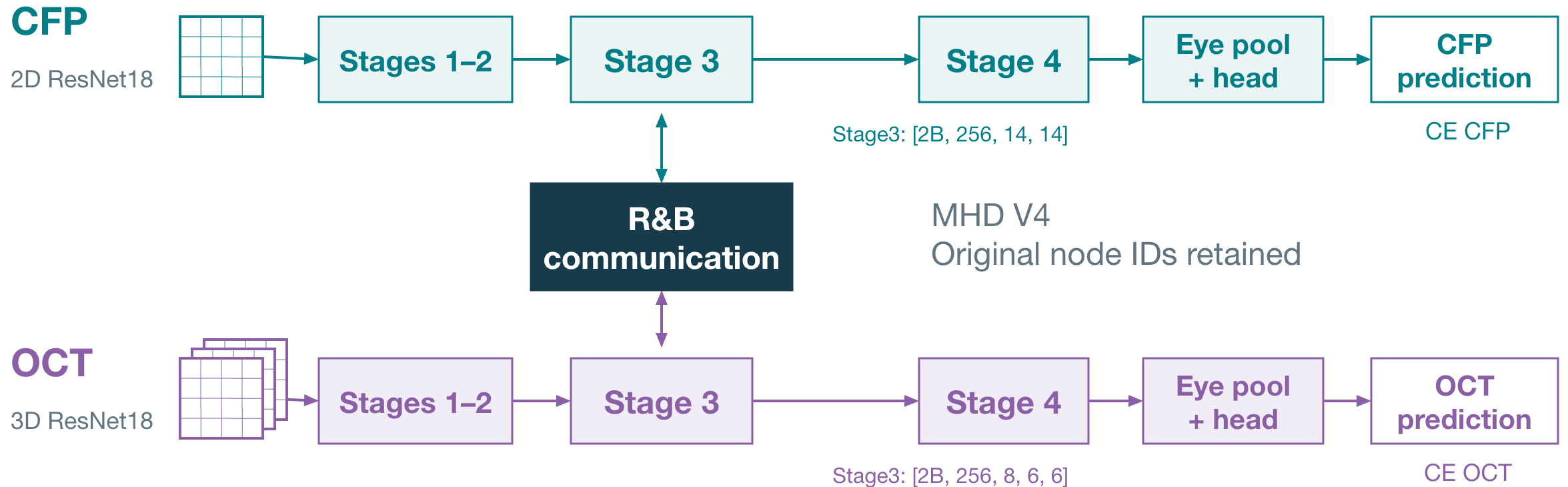
Structured communication between pretrained 2D and 3D networks



Separate predictors. Shared information.

Two independent prediction networks

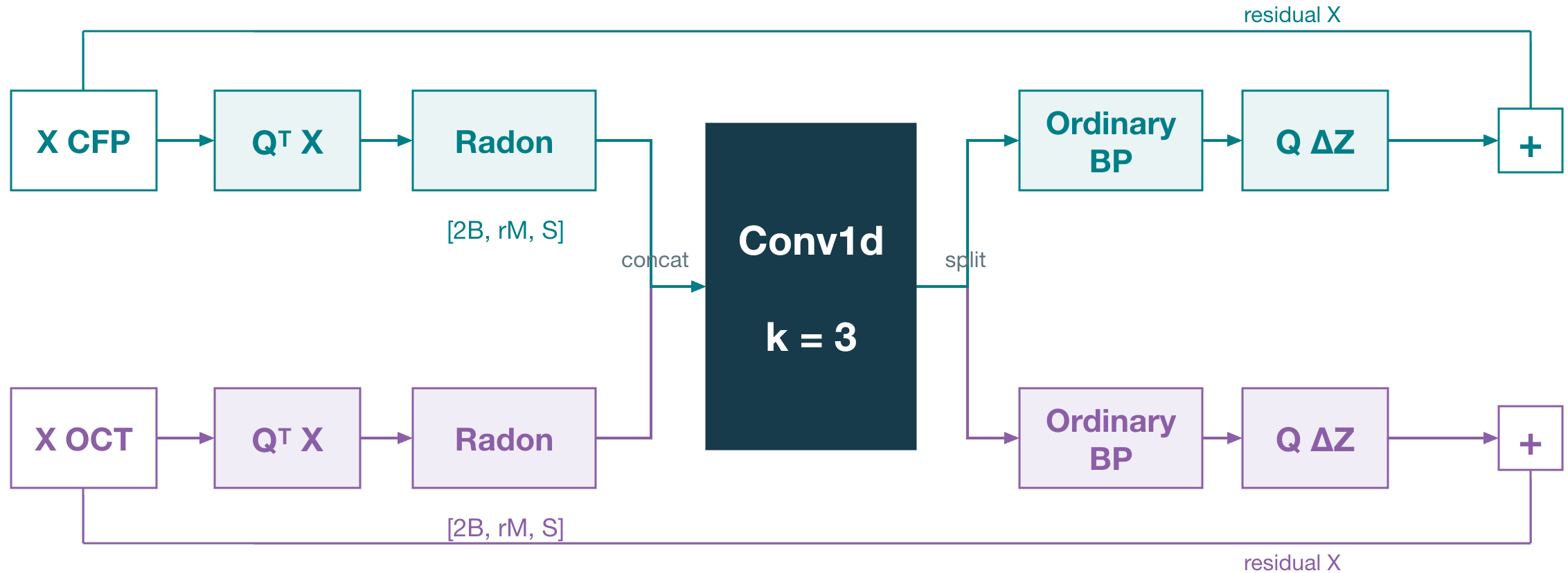
A single communication bridge connects native stage3 features



Two eyes per participant B = participants per batch

Inside a fixed-channel Radon Bridge

Fixed geometry surrounds one learnable linear communication operator



BP = ordinary backprojection

Learnable mixer

Linear residual bridge

Compression controls communication width

ρ specifies a dimension ratio, with the same meaning across widths

Learned CM projection

Radon $\rightarrow$ learned compression

After Radon

Fixed channel projection

SVD or random orthogonal Q

Before Radon

Learned channel projection

Independent encoder and decoder

Before Radon

$C = 256$ $M = 32$

ρ	r	$h = rM$
1/16	16	512
1/8	32	1024
1/4	64	2048

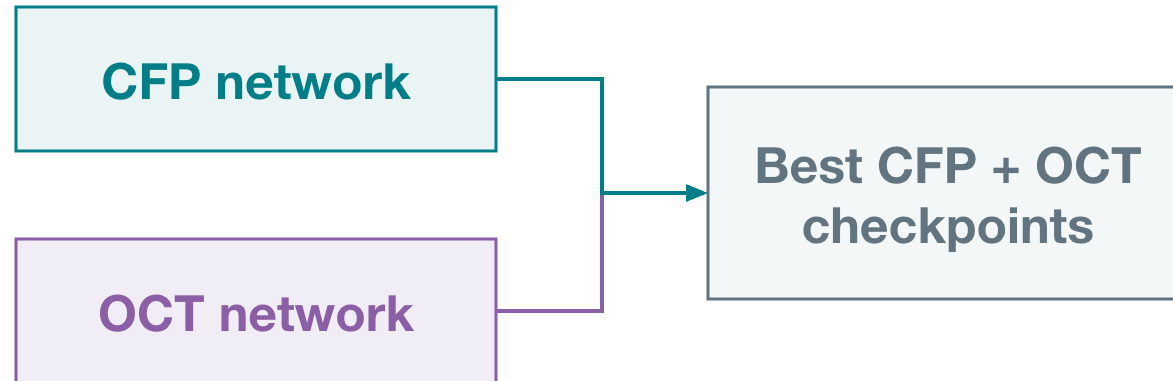
Channel ranks shown for channel projection.

Subspace reduction does not establish cross-modal semantic alignment.

Two-stage training on paired participants

UKB paired CFP / OCT: the same two networks continue into joint training

01 Independent supervision



Each branch reaches its own plateau

02 Joint finetuning

All native parameters remain trainable

1,264 training participants

296 development participants

Experiments that test the mechanism

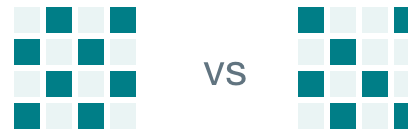
Matched interventions isolate specific explanations for a gain

Cross-source mixing



Self-only versus full mixer

Spatial structure



Radon versus scrambled / resampled

Communication direction



Bidirectional versus one direction

Correct pairing



Matched versus mismatched messages

Study map

186 completed results, followed by two separate validation studies



NEXT A

+18 QR controls

Self-only and spatial scrambling

Scheduled

NEXT B

+9 joint-width trainings

Three methods × three seeds, with three views per model

Scheduled

Research scope and evidence

A controlled benchmark of structured communication between trained networks

Core question

When does structured communication improve both native predictors?

Current evidence

Paired CFP and OCT
One record-derived glaucoma target
Three training seeds
Development-set evaluation

Independent cohorts and broader model / task validation remain future work.

Component tests first. Full results and uncertainty in the research report.